SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name: R. Jaideep

Reg No: 192525441

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Max Marks: **50 Marks**

Date : _____

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing ($\alpha = 1$). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 192525441

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	8
TOTAL	**50**	50

Total in words: fifty

Very good

50/50

Machine learning
MLA0204
Assignment-II

Name: R. Jaideep
Reg No: 192525441
Depart: AI & ML

Task 1 – Linear Regression

Given,

$x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

(a) Fit the regression equation

$$y = w_0 + w_1 x$$

using normal equations :

$$w_1 = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$w_1 = 1.99 \quad w_0 = 0.05$

$$\boxed{y = 0.05 + 1.99x}$$

(b) prediction at $x = 6$

$$y = 0.05 + 1.99(6)$$

$$\boxed{y = 11.99}$$

(c) Performance Measures

$$RSS = 0.107$$

$$MSE = \frac{0.107}{5} = 0.0214$$

$$R^2 = 1 - \frac{0.107}{40.508}$$

$$R^2 = 0.9973$$

(d) The model has a high R^2 value, indicating a good fit. However, predicting at $x = 6$ is extrapolation because the training data end at $x = 5$.

Task 2 - Ridge Regression

Given

$S_{xx} = 10, \quad S_{xy} = 19.9, \quad \lambda = 5$

Ridge slope formula:

$$w = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$w = \frac{19.9}{10+5} = 1.3267$$

Least Squares slope:

$$w_{OLS} = \frac{19.9}{10} = 1.99$$

Percentage Shrinkage:

$$= \frac{1.99 - 1.3267}{1.99} \times 100$$

$$= 33.33\%$$

Bias - variance effect: Increasing λ increases bias & decreases variance. It reduces overfitting.

Task 3 - Naive Bayes

(a) Calculate priors.

Spam = 5, Ham = 5, Total = 10.

$P(\text{Spam}) = P(\text{Ham}) = 5/10 = 0.5$

Conditional probabilities:

probability	Spam	Ham
offer=Y	4/5	1/5
offer=N	1/5	4/5
Link=Y	4/5	1/5
Link=N	1/5	4/5

(b) offer = Y, Link = Y

$$P(\text{Spam}/x) = \frac{0.5(0.8)(0.8)}{0.5(0.8)(0.8) + 0.5(0.2)(0.2)}$$

$$\boxed{P(\text{Spam}/x) = 94.12\%}$$

(c) With Laplace smoothing ($\alpha = 1$)

$$P(\text{Spam}/x) = \frac{25}{25+4}$$

$$\boxed{P(\text{Spam}/x) = 86.21\%}$$

(d) offer = N, Link = N

With smoothing

$$P(\text{Spam}/x) = \frac{4}{4+25}$$

$$\boxed{P(\text{Spam}/x) = 13.79\%}$$

Task 4:- logistic Regression

6

Given,

$x_1 = (1,2)$

$y_1 = 1$

$x_2 = (2,1)$ $y_2 = 0$.

Step1 : Initial probabilities

$$P_1 = P_2 = \sigma(0) = 0.5$$

Step2 : Gradients

$$\frac{\partial L}{\partial w_1} = 0.25$$

$$\frac{\partial L}{\partial w_2} = -0.25 \qquad \frac{\partial L}{\partial b} = 0$$

Step 3: updated parameters

$$w_1 = 0 - (0.5)(0.25)$$

$$= -0.125$$

$$w_2 = 0 - (0.5)(-0.25)$$

$$= 0.125$$

$$b = 0$$

Step 4 : New probabilities

$$P_1 = \sigma(0.125)$$

$$= 0.5312$$

Step 5 : loss

Initial loss :

$$L_{before} = 0.6931$$

Updated loss :

$$L_{after} = 0.6326$$

Task 5 - Decision Tree.

Given,

parent node = 6 Yes, 4 No.

child nodes:

a_1 : 4 Yes, 0 No

a_2 : 2 Yes, 4 No.

a) Parent entropy

$$H(S) = -0.6\log_2(0.6) - 0.4\log_2(0.4)$$

$$\boxed{H(S) = 0.971}$$

(b) Information gain

$H(a_1) = 0, \quad H(a_2) = 0.9183$

$H_{after} = 0.4(0) + 0.6(0.9183)$

$IG = 0.971 - 0.551$

$\boxed{IG = 0.420}$

(c) Gini decrease

$G(parent) = 0.48$

$G(children) = 0.2667$

$\boxed{\text{Gini decrease} = 0.2133}$

Task 6 – Cost Complexity proning

Given,

Subtree: 3 errors, 4 leaves

Single leaf: 8 errors

Total samples = 100

Step 1: critical α

$0.03 + 4\alpha = 0.08 + \alpha$

$3\alpha = 0.05$

$\alpha = 0.01667$

Step 2: for $\alpha = 0.01$

subtree cost

$0.03 + 4(0.01) = 0.07$

$0.08 + 0.01 = 0.09$

Step 3: For $\alpha = 0.02$

Subtree cost $= 0.11$

Single leaf cost $= 0.10$

Prune the subtree

Task 7: Neural Networks & Backpropagation

Given

$w_1 = 0.4, w_2 = 0.3, v = 0.5$

input $= (1, 2)$

Target $= 1$

Learning rate $= 0.5$

3

Step 1: Forward pass

$$h = \sigma(0.4 + 0.3(2))$$

$$\boxed{h = 0.7311}$$

$$\hat{y} = \sigma(0.5 \times 0.7311)$$

$$\boxed{\hat{y} = 0.5904}$$

Step 2: Initial loss

$$E = 1/2 (1 - 0.5904)^2$$

$$E = 0.0839$$

Step 3: Error terms

$$\delta_o = (1 - \hat{y})\hat{y}(1 - \hat{y})$$

$$\boxed{\delta_o = 0.09906}$$

$$S_h = S_0 \times V * h(1-h)$$

$$\boxed{S_h = 0.009738}$$

Step 4: gradients

$$\frac{\partial E}{\partial V} = -0.07242$$

$$\frac{\partial E}{\partial w_1} = -0.009738$$

$$\frac{\partial E}{\partial w_2} = -0.019476$$

Step 5: updated weights.

$$\boxed{V = 0.53621}$$

$$w_1 = 0.40487$$

$$w_2 = 0.30974$$

New loss:

$$\boxed{E_{new} = 0.08105}$$

Task 8- Support Vector Machine

Given

$$w = (0.5, 0.5) \quad b = -2$$

(a) Verify Support Vectors

Decision Function

$$f(x) = w^T x + b$$

For (3,3): $f(x) = 1$

For (1,1): $f(x) = -1$

4

Margin width $= \frac{2}{\sqrt{0.5^2 + 0.5^2}}$

$= 2.8284$

(b) classification & distances

For (4,2)

$f(x) = 1$

class $= +1$

distance $= \frac{1}{0.7071} = 1.4142$

For (1,2):

$f(x) = -0.5$

class $= -1$

distance $= \frac{0.5}{0.7071} = 0.7071$

(c) Hinge loss

$L = \max(0, 1 - y f(x))$

for (1,2), label $= -1$:

$\boxed{L = 0.5}$

for (2,2), label $= +1$

$\boxed{L = 1}$

Bothe points violate the margin because their functional margins are less than 1.

Task 9:- Ensemble learning

Given,

Individual classifier accuracy = 0.70

Error probability = 0.30.

3

(a) Three classifers

$$P = p^3 + 3p^2q$$

$$= (0.7)^3 + 3(0.7)^2(0.3)$$

$$= 78.4\%$$

(b) Five classifiers

$$P = 10p^3q^2 + 5p^2q + p^5$$

$$\boxed{P = 83.692\%}$$

Task 10 - AdaBoost

given,

Five samples with equal weights.

Samples 2 & 5 are misclassified.

4

Initial weight:

$$w_i = 1/5 = 0.2$$

Step 1: weighted error

$$\varepsilon = 0.2 + 0.2$$

$$\boxed{\varepsilon = 0.4}$$

Step 2 : learner weight

$$\alpha = 1/2 \ln \frac{1-0.4}{0.4}$$

$$\boxed{\alpha = 0.2027}$$

Step 3 : unnormalised weights.

Correctly classified

$$0.2e^{-0.2027} = 0.1633$$

Misclassified :

$$0.2e^{0.027} = 0.2499$$

Step 4 : Normalised weights.

Sample	weight
1.	0.1667
2.	0.2500
3.	0.1667
4.	0.1667
5.	0.2500.

Task II - Linear Discriminant Analysis

given

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$\mu_1 = (1,2) \qquad \mu_2 = (3,1)$$

(a) Discriminant direction

$$w = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$\Sigma^{-1} = \frac{1}{1.75}\begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$$

$$\boxed{w = (1.4286, -1.7143)}$$

Equal-prior threshold:

$$c = \tfrac{1}{2} w^T (\mu_1 + \mu_2)$$

$$\boxed{c = 0.2857}$$

(b) classify $x = (2.5, 1)$

$$w^T x = ((1.4286)(2.5) - 1.7143$$

$$= 1.8571$$

$$1.8571 > 0.857$$

$$\boxed{\text{class} = \omega_2}$$

(c) unequal priors

$$P(\omega_1) = 0.8, \quad P(\omega_2) = 0.2$$

$$c' = 0.2857 - \ln(0.2/0.8)$$

$$\boxed{c' = 1.6720}$$

$$1.8571 > 1.6720$$

class remains : ω_2

Task 12 - classifier Evaluation.

Given 1000 test loans and 100 actual defaulters

FN Cost = ₹ 50,000

FP cost = ₹ 5,000.

(a) Performance metrics

$$\text{Accuracy} = \frac{TP + TN}{1000} \times 100$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93
Precision	0.4375	0.3208	0.6667
Recall	0.70	0.85	0.60
F1-Score	0.5385	0.4658	0.6316

(b) Expected Cost

Cost = (FN × 50000) + (FP × 5000)

Model A = 30(50000) + 90(5000)

Model A = 19,50,000

Model B = 15(50000) + 180(5000)

Model B = 16,50,000

Model C = 40(50000) + 30(5000)

= ₹ 21,50,000

(c) Select the best model

Model B is the best choise based on financial cost

Reasons:

→ It has the lowest expected cost of ₹18,50,000

→ It has the highest recall of 85%

→ It misses only 15 defaulters.

(d) Improvement and validation.

1) use cost-sensitive learning to give more importance to defaulters.

2) Adjust the classification threshold to reduce false negatives & and total cost.

Validation: use stratified cross-validation & an independent test dataset to prevent overfitting.